

Findings from Instagram Likes Prediction Using Machine Learning

Model Performance Overview

The project evaluated various machine learning models using both traditional features and CLIP-based image embeddings to predict Instagram likes. Below are the tables summarizing the accuracy and error metrics for classification and regression tasks.

1. Classification Task

Model	Accuracy (%)
RandomForestClassifier (Traditional Features)	95.80
XGBoost (With CLIP Features)	97.20
XGBoost (With Selected CLIP Features)	97.62
XGBoost with PCA	79.58
CatBoost Classifier	96.92

2. Regression Task

Model	Mean Absolute Error (MAE)
Linear Regression (Traditional Features)	97,272.96
XGBoost (With CLIP Features)	81,443.11

Key Findings

- Best Classifier:** The **XGBoost (With Selected CLIP Features)** achieved the highest accuracy of 97.62%, proving that selecting the most important CLIP features significantly enhances performance.
- Regression Improvement:** Using CLIP features reduced the Mean Absolute Error from 97,272.96 to 81,443.11, demonstrating the importance of image content in predicting likes.

Suggestions for Further Improvement

- Hyperparameter Tuning:** Implement Bayesian Optimization for further tuning.
- Incorporate Textual Data:** Analyze captions/hashtags to extract additional engagement signals.
- Deep Learning Models:** Experiment with CNNs or transformers for richer feature extraction beyond clip.